Supplementary Information: Anomalous Nuclear Quantum Effects in Ice

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BASIS SET

We used a basis set of triple-ζ polarized orbitals in all calculations. The SIESTA basis set input file, with information about the cutoff radii of the orbitals is given below. We present the same information for double-ζ polarized orbital basis set we used in previous studies [1]. In order to compare the performance of this basis set, we calculated the lattice parameters of the proton ordered (H-ordered) ice XI structure at Γ point for the PBE functional. The first two lines of Table I show that a triple-ζ polarized orbital basis improves the results as compared to plane wave calculations [2].

Triple-ζ basis:

```
%Block PAO.Basis
O 3 -0.18361
  n=2 0 3 E 50.06241 5.0
      7.0 4.0 2.4
      1.00000 1.00000 1.0
  n=2 1 3 E 10.0 6.0
      7.0 4.0 2.2
      1.00000 1.00000 1.0
  n=3 2 1 E 50.0 0.0
      6.0
      1.00000
H 2 0.94703
  n=1 0 3 E 50.0 6.0
      7.0 4.0 2.0
      1.00000 1.00000 1.0
  n=2 1 1 E 1000.0 0.0
      6.0
      1.00000
%EndBlock PAO.Basis
```

Double-ζ basis:

```
%Block PAO.Basis
O 3 -0.24233
  n=2 0 2 E 23.36061 3.39721
      4.50769 2.64066
      1.00000 1.00000
  n=2 1 2 E 2.78334 5.14253
      6.14996 2.59356
      1.00000 1.00000
  n=3 2 1 E 63.98188 0.16104
      3.54403
      1.00000
H 2 0.46527
  n=1 0 2 E 99.93138 2.59932
      4.20357 1.84463
      1.00000 1.00000
  n=2 1 1 E 24.56504 2.20231
      3.52816
      1.00000
%EndBlock PAO.Basis
```

TABLE I. a and c lattice parameters, their ratio (c/a), corresponding volume per molecule of H-ordered and H-disordered ice Ih, and the cohesive energy (E_c) of the H-ordered ice calculated for several DFT functionals. H-ordered ice XI with PBE functional is calculated for double-ζ polarized (DZP) and triple-ζ polarized (TZP) atomic orbital basis. H-disordered ice Ih with the PBE functional has the same lattice parameter as the H-ordered ice Ih, whereas H-disordered system with the vdW-DF^{PBE} has a larger volume than the H-ordered structure. Structural experimental results are for $T = 10$ K. All lengths and volumes are in Å and Å^3, and the cohesive energy is in eV.

H-ordered	a	c	c/a	Vol/H_2O	E_c
PBE (DZP)	4.44	7.25	1.633	30.94	0.627
PBE (TZP)	4.39	7.17	1.633	29.92	0.666
PBE (BF)	4.39	7.17	1.633	29.92	0.666
PBE [2]	4.40	7.20	1.636	30.23	0.665
vdW-DF^{PBE}	4.44	7.23	1.628	30.86	0.770
revPBE	4.52	7.42	1.642	32.82	0.481
revPBE [2]	4.53	7.45	1.642	33.13	0.476
vdW-DF^{revPBE}	4.56	7.43	1.629	33.45	0.590
H-disordered	a	c	c/a	Vol/H_2O	
PBE	4.39	7.17	1.633	29.92	
vdW-DF^{PBE}	4.44	7.24	1.630	30.90	
Expt. H_2O [3]	4.497	7.321	1.628	32.05	0.61[4]
Expt. D_2O [3]	4.498	7.324	1.628	32.08	

STRUCTURE

The structure of the H-ordered systems (i) ice XI with 4 molecules in the unit cell (Fig. 1) and (ii) Bernal-Fowler (BF) ice with 12 molecules in the unit cell (Fig. 2) are compared for the PBE functional. The results in Table I show that ice XI and BF have the same lattice parameters for the same unit cell.

The H-disordered ice Ih structure has 96 molecules in the $3a \times 2\sqrt{3}a \times 2c$ cell (Fig. 3). The a and c parameters given in Table I correspond to the unit cell parameters of this system.

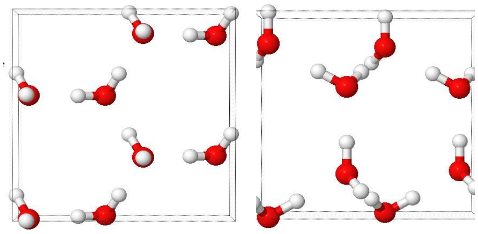

FIG. 1. Unit cell of the H-ordered ice XI structure. The image on the right is the top view of the x-y plane; the image on the left is the side view of the x-z plane.

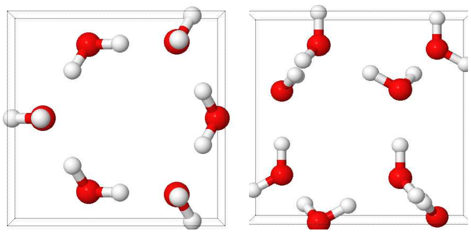

FIG. 2. Bernal-Fowler ice Ih structure. The image on the right is the top view of the x-y plane; the image on the left is the side view of the x-z plane.

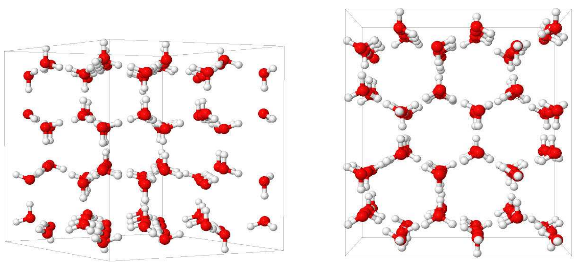

FIG. 3. H-disordered ice Ih structure. The image on the right is the top view of the x-y plane; the image on the left is the side view of the x-z plane.

PHONON CALCULATIONS

All the relaxations are performed using a real-space mesh cutoff of 500 Ry for the real space integrals, electronic k-grid cutoff of 10 Å (corresponding to 38 k-points) for unit cell calculations, force tolerance of 0.001 eV/Å and a density matrix tolerance of 10^{-5} electrons.

The force constant calculations of H-ordered ice XI structure are performed with a finer real-space mesh cutoff of 800 Ry. We used an atomic displacement Δx=0.06 Å for the frozen phonon calculation.

The force constant calculations of H-disordered ice Ih are performed with a mesh cutoff of 500 Ry. We used an atomic displacement Δx=0.08 Å for the frozen phonon calculation.

We tested these parameters to obtain force constants in phonon calculations as accurate as possible, so that the Grüneisen parameter calculations will have minimum noise. Results of our error estimation are presented below.

Frozen Volume

Instead of doing a variable cell optimization, for each functional, we optimize the coordinates as a function of the volume to calculate the ground state frozen volume and lattice parameters. The curve for Kohn-Sham energy as a function of volume is obtained and fitted to a third order polynomial. The minimum of the energy-volume curve is the frozen volume of the system used in the free energy calculations with the quasi-harmonic approximation.

QUASI-HARMONIC APPROXIMATION

We present a brief derivation of the frequency as a function of volume within the quasi-harmonic approximation (QHA). Starting with the definition of Grüneisen parameters given in eq.(1)

$$\gamma_k = -\frac{\partial(\ln\omega_k)}{\partial(\ln V)} = -\frac{V}{\omega_k}\frac{\partial\omega_k}{\partial V} \tag{1}$$

$$\int_{\omega(V_0)}^{\omega_k} \frac{\partial\omega_k}{\omega_k} = -\int_{V_0}^{V} \gamma_k \frac{\partial V}{V} \tag{2}$$

$$\ln\left(\frac{\omega_k}{\omega(V_0)}\right) = -\gamma_k \ln\left(\frac{V}{V_0}\right) \tag{3}$$

This is the quasi-harmonic approximation QH1:

$$\omega_k(V) = \omega(V_0)\left(\frac{V}{V_0}\right)^{-\gamma_k} \tag{4}$$

If the Taylor expansion of the logarithm in eq.(3) is taken instead, then it gives QH2:

$$\omega_k(V) = \omega(V_0)\left(1 - \gamma_k \frac{V - V_0}{V_0}\right) \tag{5}$$

As seen in Fig 4, QH2 performs better than QH1. Indeed it is an excellent approximation to the full QHA, coined QH0 in the figure, in which the frequencies are calculated at each volume. Using DFT instead of the empirical potential the differences between QH1 and QH2 are much smaller, as seen in the error estimation section.

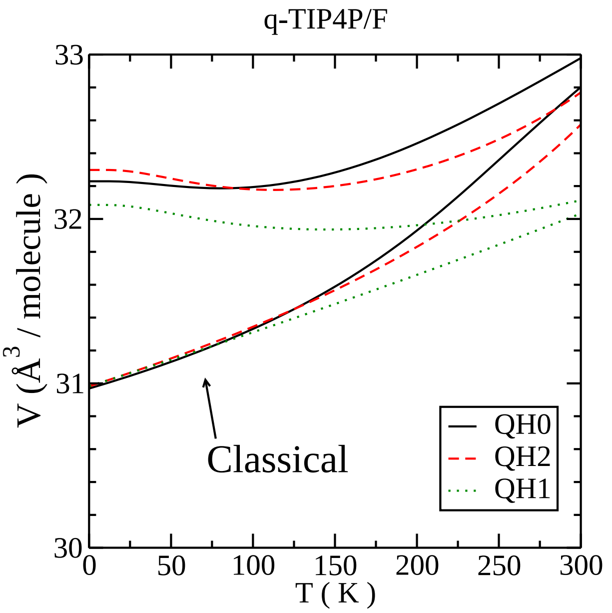

FIG. 4. (Color online). Volume per molecule for H_2O and for the classical limit calculated using q-TIP4P/F using different versions of the QHA. QH0 stands for the full QHA, QH1 and QH2 are explained in the text.

In order to calculate the Grüneisen parameter in eq.(1), we calculated the frequencies for different volumes obtained by changing the lattice parameters by 0.15% around the equilibrium volume. Frequencies of 5 different volumes are calculated for the calculations where the phonons are computed only for Γ sampling. 3 volumes are used for when the full Brillouin zone is sampled for the phonon calculation.

Volume Change at T = 0

The quasi-harmonic volume shift is:

$$\frac{V - V_0}{V_0} = \frac{1}{V_0 B_0} \sum_k \gamma_k \hbar\omega_k \left(n_k + \frac{1}{2} \right) \tag{6}$$

where V_0 is the frozen or classical volume, V is the quantum volume and B_0 is the bulk modulus of the system.

At $T = 0$, $n_k = 0$ and it can be seen that $\Delta V = V - V_0$ is proportional to the average $\langle \gamma_k \omega_k \rangle$. The anomalous isotope shift observed when replacing H by D implies that $\Delta V(H_2O) < \Delta V(D_2O)$. Figure 5 shows $< \gamma_k \omega_k >$ for each branch as a function of the frequency for H_2O and D_2O. For both molecules, the translation and libration branches, less energetic than the OH stretching branches, have positive Grüneisen constants. The bending modes have a Grüneisen constant very close to 0, so they contribute very little to the sum. The stretching modes of the covalent OH bond have negative Grüneisen constants and larger frequency. Therefore, it is clear that in ice Ih, the isotope shift depends on a fine balance between the contribution of the H-bond-related frequencies with $\gamma_k > 0$ and branches related to the OH covalent bond with $\gamma_k < 0$.

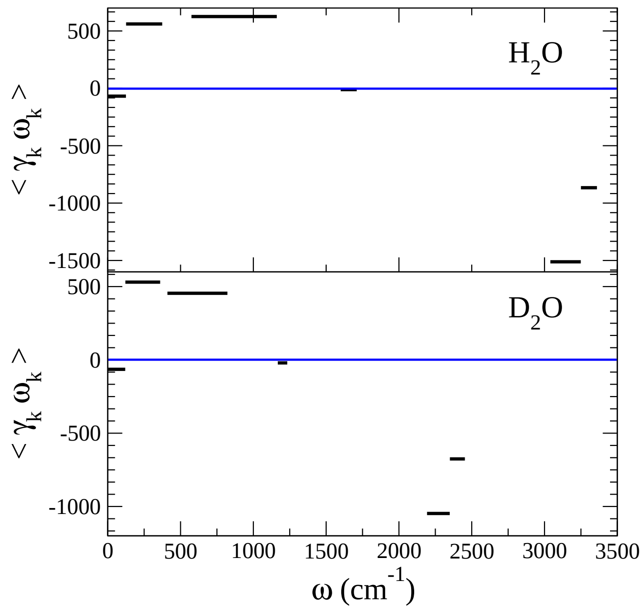

FIG. 5. $\langle \gamma_k \omega_k \rangle$ for each phonon branch calculated using vdW-DF^{PBE}. Top: H_2O, bottom: D_2O. The continuous horizontal line indicates $\langle \gamma_k \omega_k \rangle = 0$

ERROR ESTIMATION

We present here an analysis of the main numerical sources of error in our calculations, beyond any DFT approximation. We have identified three:

Numerical evaluation of Grüneisen constants:

We choose 5 volumes slightly below and above V_0 to calculate ω_k and γ_k for the calculations where the phonons are obtained for Γ-sampling only. However, we choose 3 volumes when the QHA is computed using phonons obtained by sampling the full (4 molecules, hexagonal) Brillouin zone. To estimate the error involved in calculating Grüneisen constants from 3 or 5 volumes we compare the vdW-DF^{PBE} functional volume QHA results obtained using Γ-sampling phonons. As seen in Table II, the volume decreases by ~0.03% when 5 volumes are used instead of 3. The magnitude of the isotope shift is hardly affected.

TABLE II. Isotope-dependent volume per molecule (Å) at T=0 for H-ordered ice XI structure with vdW-DF^{PBE} functional at Γ point. *cla* represents the classical result. (i) shows the V(T=0) as function of the number of volumes chosen to calculate ω_k and γ_k. (ii) As function of the Δx atomic displacement used to compute the force constants. (iii) As function of the two different QHA methods, QH1 to QH2.

(i) # of volumes	5	3
cla	30.88	30.88
H_2O	31.01	31.02
D_2O	31.10	31.11
$H_2{}^{18}O$	30.98	30.99
(ii) Δx for phonons	0.06 Å	0.08 Å
cla	30.88	30.88
H_2O	31.02	30.97
D_2O	31.11	31.07
$H_2{}^{18}O$	30.99	30.95
(iii) QHA method	QH1	QH2
cla	30.88	30.88
H_2O	31.00	31.01
D_2O	31.09	31.10
$H_2{}^{18}O$	30.97	30.98

Frozen phonon approximation:

The second possible source of error is the atomic displacement Δx used to compute the force constants by finite differences, within the frozen phonon approximation. We compare vdW-DF^{PBE} volumes at Γ with mesh cutoff of 800 Ry using 3 volumes to compute ω_k and γ_k. Phonons are calculated using two different displacements: $\Delta x = 0.06$ Å and $\Delta x = 0.08$ Å. Table II shows that this is the largest source of error. Volumes for the same isotope are modified $\sim 0.14\%$ when $\Delta x = 0.08$ Å changes to $\Delta x = 0.06$ Å. However, the volume isotope shift is almost unchanged.

Numerical quasiharmonic approximation:

The last source of error is due to the method of linearizing the volume dependence of the frequencies in the quasiharmonic approximation. This has already been discussed above, using the q-TIP4P/F model, but we present results using DFT in this section. We compare results for the QH1 and QH2 linearizations using the vdW-DF^{PBE} functional (Γ-sampling for phonons). The real-space mesh cutoff is 800 Ry and the atomic displacement Δx=0.06 Å for force constant calculations. We used 5 different volumes to obtain the frequencies and Grüneisen constants. As seen in Table. II QH1 predicts smaller volumes than QH2, with a difference of ∼0.03%. This is much smaller than the differences obtained using the q-TIP4P/F force field. As expected, both give the same sign and almost no change for the isotope effect. Overall, as seen above, QH2 gives better results compared to the experiments.

TEMPERATURE DEPENDENT ISOTOPE VOLUME FOR PBE

We present in Fig. 6 the same Fig. 2 of the main paper, but computed for PBE instead of vdW-DF^{PBE}. As PBE predicts a stronger covalent-Hbond anti-correlation effect, the anomalous effect is overestimated in this functional. This overestimation translates into a smaller (0.5%) contraction of the frozen (classical) volume at $T = 0$, as compared to $\sim 1\%$ obtained with vdW-DF^{PBE}. Also, the net quantum effect becomes anomalous at $T \sim 30$ K, lower than the transition predicted by vdW-DF^{PBE} ($T \sim 70$ K).

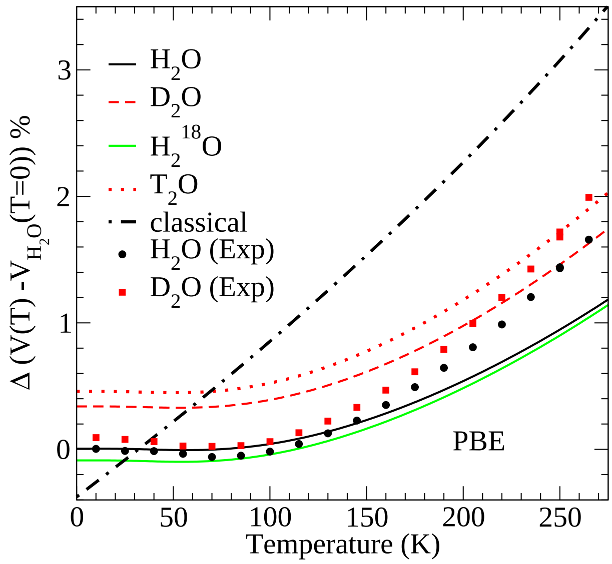

FIG. 6. (Color Online). Volume change $V(T)/V_{H_2O}(0) - 1$, relative to that of H_2O at $T = 0$, for different isotopes calculated using the QHA with the PBE functional. Also shown are the experimental results from Ref. 3.

ANOMALOUS ISOTOPE EFFECT FOR VDW-DF^{revPBE}

As mentioned in the text, vdW-DF^{revPBE} does not reproduce the T=0K inverse isotope effect between H_2O and D_2O. Here we show that the anomalous effect is still predicted but for T>∼100K. The behavior is still very different from the q-TIP4P/F classical potential, in the sense that the latter never predicts an anomalous isotope shift in the volume. According to this plot, the anomalous shift by this functional is also important at the melting temperature and therefore might be relevant for the liquid phase.

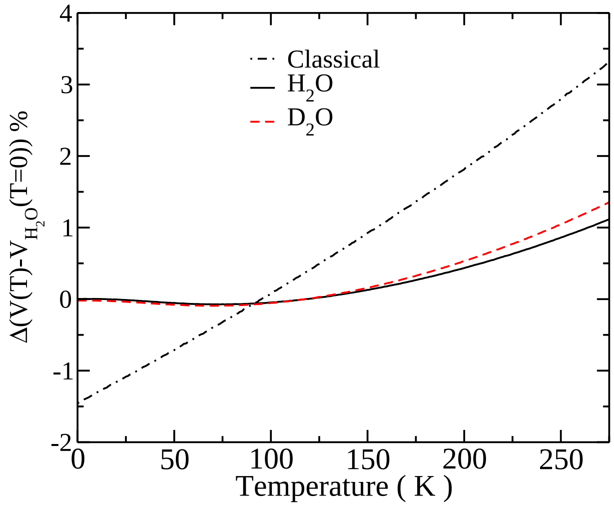

FIG. 7. (Color Online). Volume change $V(T)/V_{H_2O}(0) - 1$, relative to that of H_2O at $T = 0$, for different H_2O, D_2O and the classical limit calculated using the QHA with the vdW-DF^{revPBE} functional.

ANOMALOUS ISOTOPE EFFECT FOR TTM3-F

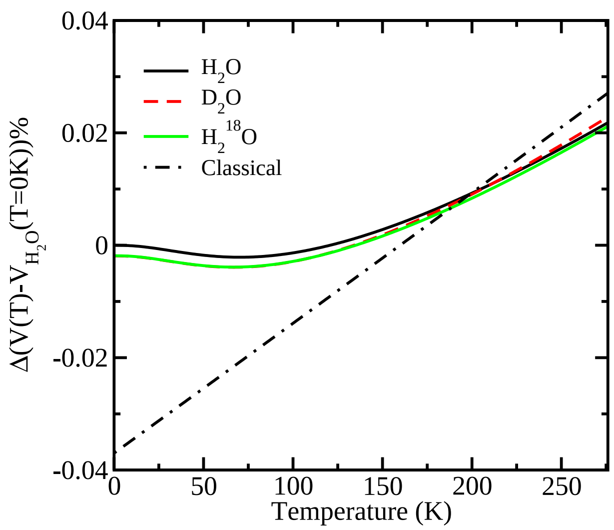

FIG. 8. (Color Online). Volume change $V(T)/V_{H_2O}(0) - 1$, relative to that of H_2O at $T = 0$, for different H_2O, D_2O, $H_2{}^{18}O$ and the classical limit calculated using the QHA with TTM3-F force field model.

We have repeated the QHA for TTM3-F[5] using the same 96 molecules cell with disordered proton structure as described in the text. The frozen phonon approximation was used to compute the phonons at three different volumes. We have used both a finite atom displacement of 0.02 Å and 0.04 Å, both of which provided very close results. Grüneisen constants were computed numerically from these three frequency sets, as described previously.

The T=0 isotope effect is not reproduced using the TTM3-F[5] polarizable force field. The absolute values are given in the table in the text, and the relative values are shown in Fig. 8. Due to a cancellation of competing anharmonicities, it is very difficult to distinguish between the the isotopes with this model. Indeed, as it can be seen the overall thermal expansion is tiny, less than 0.02% at the melting temperature. At T∼230K, the volumes of H_2O and D_2O converge and only close to melting temperature the anomalous isotope shift is predicted. This behavior is similar to that of vdW-DF^{revPBE}, but the anticorrelation between covalent and H-bonds is still too weak to predict a correct isotope effect.

In ref [6], the authors studied of vibrational properties of TTM3-F in the liquid phase using PIMD simulations. They concluded that in quantum simulations the infrared spectrum of TTM3-F fails to reproduce experiments and they point to the need of refinement of the model in order to improve its vibrational description. We believe that this improvement will also allow for a better description of the anomalous isotope effect.

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